LS-CAT Management Board Meeting

Keith Brister

12 Noon Central Time 1 PM Eastern Time

Call In #: 1-888-850-4523
Passcode: 190580

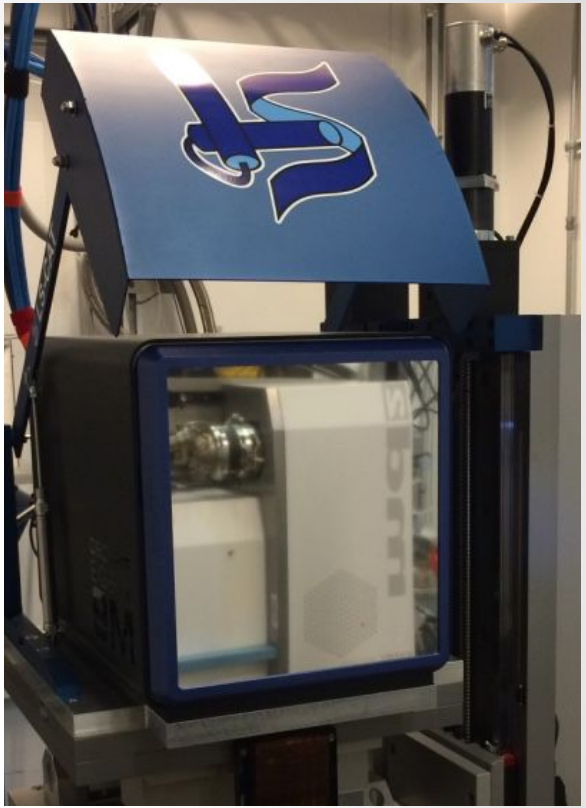
http://conf1.ls-cat.org
http://conf2.ls-cat.org

Agenda

1. Eiger Commissioning
2. Scheduling
3. APS-U Status
4. LS-CAT Extension
5. FY 2017 Budget (Votes Needed)
6. New Business

Eiger 9M Installed in 21-ID-D!

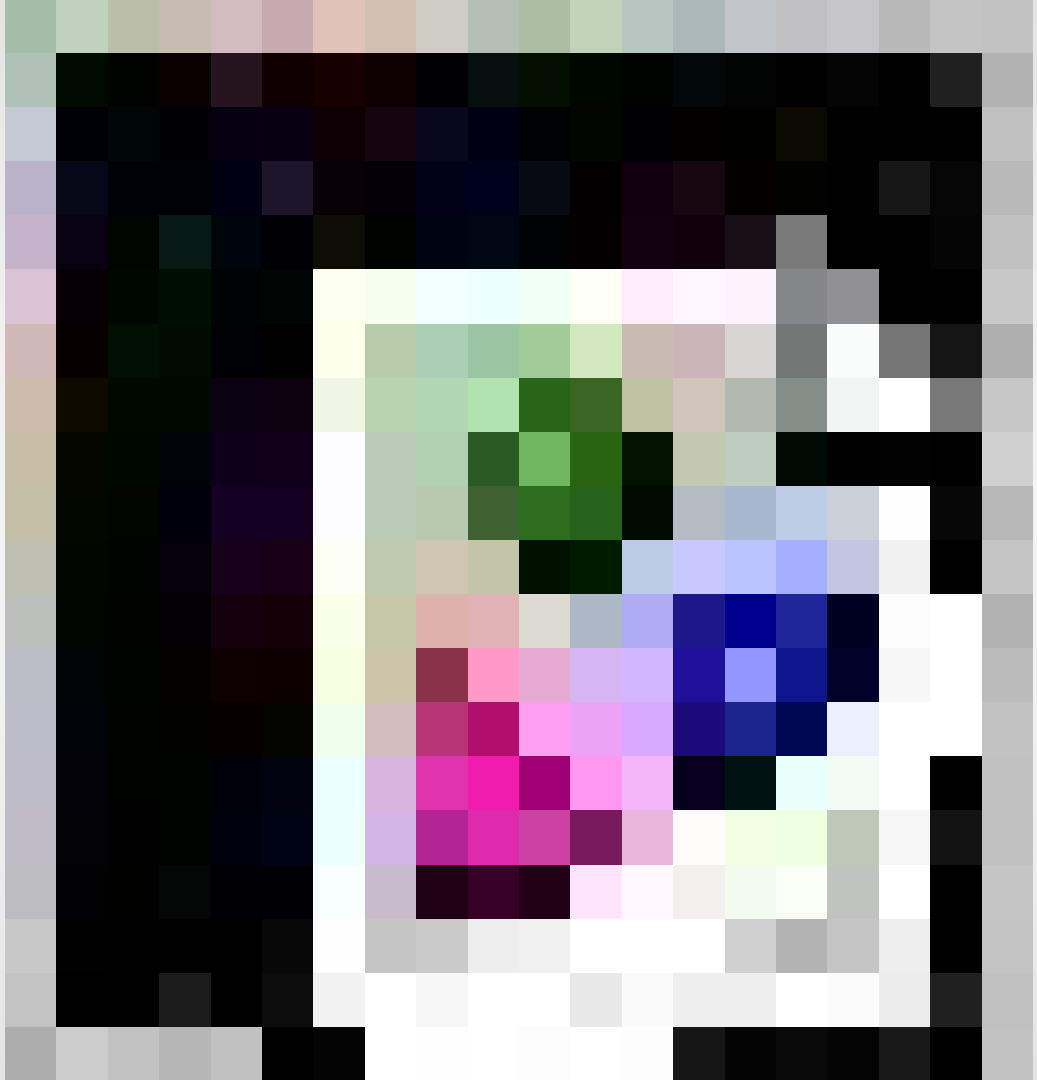




It’s alive!

Most common response upon seeing the data being collected: “Wow!”

1° / Sec
0.5° / Frame
12.0 KeV
150 mm Distance

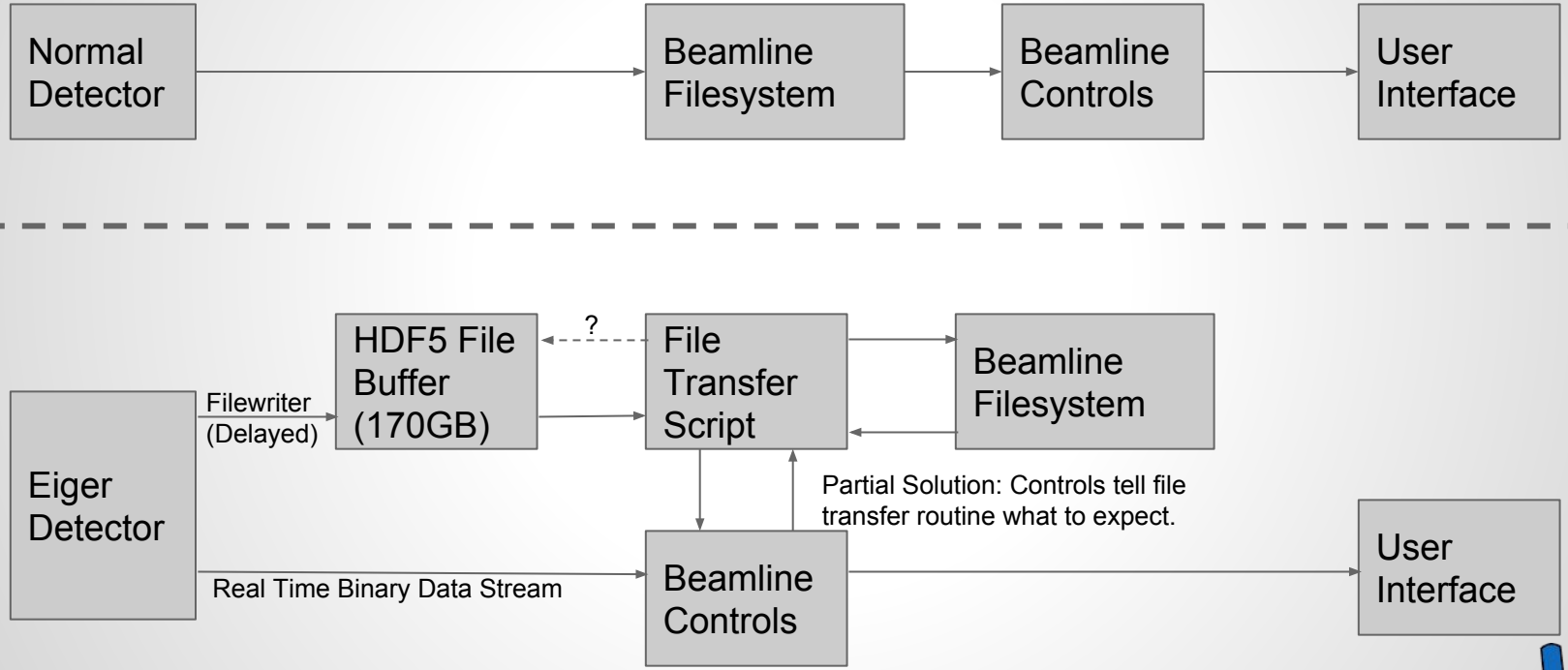


Some Commissioning Challenges

- Calculation for the width of the pulse that opens the shutter (and triggers the detector) was incorrect leading to problems getting data collection to start early in commissioning.
- HDF5 format files are not (yet) well accepted by the community.
- Temporary bad lines and/or modules (Mainly during the last week of the run, fixed by dectris firmware update).
- Two failed modules. Fixed by Dectris technician.
- Detector too well centered in beam: detector gaps reduce completeness. (We'll move the detector to a less symmetric position and advise users to collect a second dataset at a different distance if they suspect they have a low symmetry space group).

And more in following slides...

There is no way to reliably read data from the detector.



Remote Users Are Flying Blind

- To offer something, streaming images from the detector are converted to a “webcam” format and displayed as such. While better than nothing, the inability to change the contrast and view arbitrary frames (not to mention the lack of resolution rings) made using the interface remotely frustrating.
- Local users were not so limited.
- This should be fixed early in 2016-2 if not at the beginning of 2016-2.

This Detector Spews Data

- While datasets are typically 20GB in size (only about 3 or 4 times those of the Rayonix 300) it's possible to collect many more datasets in any given amount of time. It's also possible to collected datasets that are much larger than 20GB.
- With a spindle rate of 10 degrees per second you can collect 180 degrees of data in 18 seconds, about the time it takes to collect 7 frames on the Rayonics detectors. People are getting more comfortable removing the attenuation and turning the spindle faster and collecting more data.
- We've added support for Globus Connect to help move the data but it will take another run or two for our users (and our staff) to make optimal use of this method of file transfer.
- Our users will likely need to invest in the following:
    - Disk space to receive the data.
    - Strategies to filter out garbage, ie, don't collect useless data and/or delete forever data you cannot use.
- We are adapting our archive strategy.

The problems we are encountering are typical of commissioning any new detector and will be worked out with time.

All told, this is an exciting new era of data collection at LS-CAT!

Scheduling

We have been allowing members to schedule their time mainly with 10AM to 10PM slots. Practically this means that most of the remaining time for Rapid Access and for General Users is now between 10PM and 10AM. For experienced users this is not a problem but for new users or users that require a fair amount of help this is putting a strain on the staff.

The bottom line: At least some of your allocation will start coming in the form of locally collected night shifts.

APS-U

The latest expected start date for the long down period is 2022. This one year delay is do to the lack of an increase of funds for the APS-U project. (Attributed, oddly, to the climate change agreement.)

The next milestone is the “baselining” where the scope and schedule for the project will be defined. Expect this to be very conservative with many useful projects being placed in contingency.

It’s just about time for the LS-CAT members to decide what we should be doing to prepare for an upgraded APS, mainly, where is the money for an LS-CAT upgrade going to come from. Expect something on the order of $2M per share to bring the facility up to date. $4M per share if you want to be pushing the envelope (whatever that means).

LS-CAT 5 Year Extension

Current agreement expires February 28, 2017.

The amendment that extends the current agreement to February 28, 2022 has been signed by:

- University of Michigan
- University of Wisconsin
- Vanderbilt University
- University of Illinois

(Thanks!)

Need to sign next:

- Michigan State University
- Van Andel University
- Wayne State University

Northwestern University will sign last and send copies of the signing sheet to the others.

Budget Details

FY 16 Spending vs FY 16 Budget

FY 17 with 3% increase.

Description	FY16 Total (Est)	FY2016 Budget	FY 17 3%
Salaries	884,980	882,579	911,053
Fringe Benefits	240,715	241,826	252,362
Supplies	57,602	36,000	40,000
Supplies, Restricted	1,467	2,500	2,500
Services	54,062	27,000	30,000
Services, Restricted	18,897	18,000	20,000
External Space Rental	120,161	120,161	123,766
Travel, Domestic	2,311	5,000	5,000
Utilities & Telephone	8,901	8,000	9,000
Capital Equipment	153,586	109,348	96,607
Indirect Cost - Sponsored	318,782	379,639	394,666
Total	**1,861,464**	**1,830,053**	**1,884,955**

Budget FY 2017

Salaries			
Base	911,053		100% KB, JSB, MB, JVO, EK, DS, SA, ZW
Fringe	252,362		95% NB
Total		1,163,415	
Items w/o F&A			
Capital Eq.	96,607		
Rent	123,766		
Telcom	9,000		
Total		229,373	
Items w/F&A			
Lab Supplies	40,000		
Office Supplies	2,500		
Lab Services	30,000		
Insurance	20,000		
Travel	5,000		
Total		97,500	
F&A		394,666	
Total		1,884,954	
Per Share		327,818	⇐ NU, UM, MSU, VARI
Per 0.5 Share		163,909	⇐ UI, VU, UW
Per 0.25 Share		81,955	⇐ WSU

Vote 1:
Approval of amounts shown.

Vote 2:
Exact amount of capital equipment needed depends on what breaks and what upgrades cannot be postponed. Monies left on this line will be added to capital fund (currently ~$800K) or, conversely, when outlays exceed this amount we'll dip into the capital fund. As always, the board will be kept informed about unusually large expenses. Vote 2 allows the capital equipment funds to be used for normal upkeep of the facility.